# Agentic AI in mathematical research

# A brief disclaimer

I am not an expert in this field. I have been exploring it out of curiosity and to stay up to date.

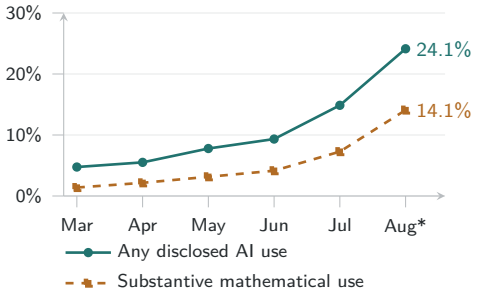
This talk neither advocates nor argues against using AI to write papers.

- A brief history and overview of mathematical research with LLMs, focusing on agentic approaches.
- What the literature tells us about successful approaches and their limitations.
- A case study where I build an AI research team to write a paper.

# AI use and mathematical output

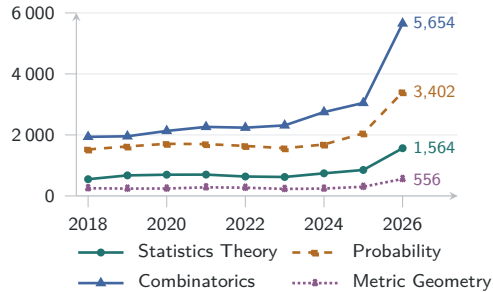
## AI disclosures on arXiv

Share of math-related submissions, 2026




## Submissions by field

New preprints, January–August each year

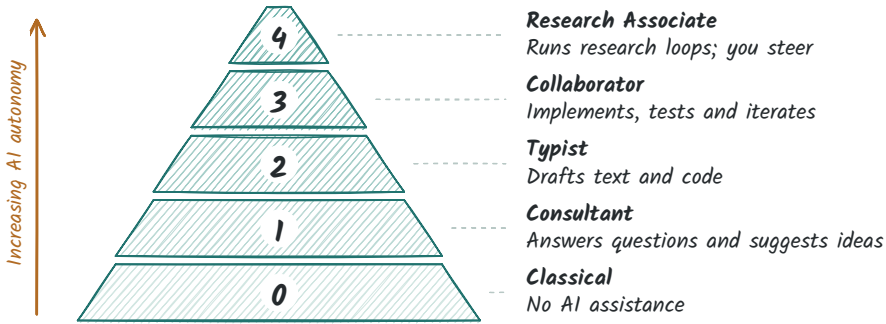



Left: 3,575 of 32,944 preprints disclosed AI use. *August covers 1–20.
Right: primary-category counts. Growth alone does not identify an AI effect.

Jin, Ke & Sui (2026), Table 1 **Explore arXiv Math Pulse** GitHub source.

# How have you used AI in research?

Which is the highest level you have used?




Adapted from Zimmer et al. (2026), *The Agentic Researcher*, Section 2.

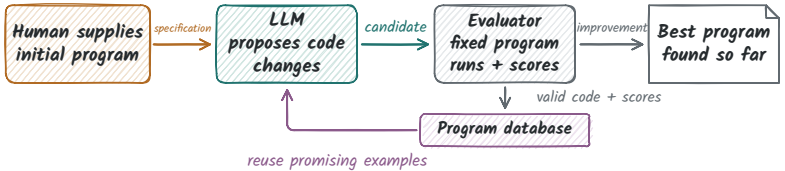
# A short history of math research with LLMs

## Agentic research systems over time

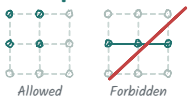
A brief history, with a focus on mathematics.

# FunSearch Romera-Paredes et al., *Nature* (2024)

An evolutionary approach with fixed evaluation criteria implemented in computer programs.




## Larger cap sets



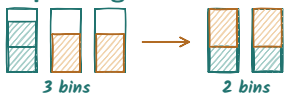

496 → 512

points in 8 dimensions

No three distinct points on a line in $\mathbb{F}_3^n$.

A new construction, not a proof of maximality.

## Online bin-packing heuristics



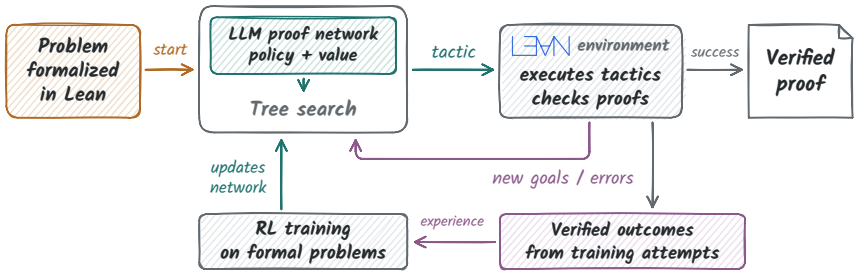

Use as few fixed-capacity bins as possible; place each item on arrival.

**AlphaEvolve (2025)** later extended this approach: $4 \times 4$ complex matrix multiplication with **48 scalar multiplications**, versus Strassen's 49.

Workflow: Fig. 1; results: Fig. 4 and Table 1. Romera-Paredes et al. (2024).

# AlphaProof Hubert et al., *Nature* (2026)

Another way to verify and evaluate mathematics is through formal verification in Lean.




**Search:** the LLM proposes tactics; Lean returns new proof states.

**Learning:** The model learns from proof attempts. For harder problems, it practises on similar ones.

**IMO 2024: silver-medal level — 28/42 points with AlphaGeometry 2.**

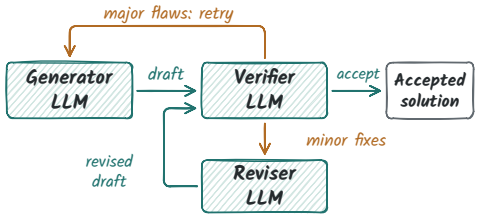
AlphaProof solved 3 problems; AlphaGeometry 2 solved 1.

Adapted from Figs. 1–2 and the AlphaProof section. *Nature* **651**, 607–613 (2026).

# Aletheia Feng et al., arXiv (2026)

We can also use an independent LLM for verification. This does not guarantee correctness.

## Gemini Deep Think agents




LLM judgment; no formal guarantee (Fig. 1).

## Reported mathematical results

- Erdős: 2 full + 2 partial new solutions from 700 problems, assessed by experts.

And other mathematical achievements.

## Other LLM research systems

**The AI Scientist** *Nature* (2026)
Ran ML experiments and wrote a paper that passed a workshop's first review round.

**Google Co-Scientist** *Nature* (2026)
Agents proposed and refined leukaemia drug combinations, validated in laboratory experiments.

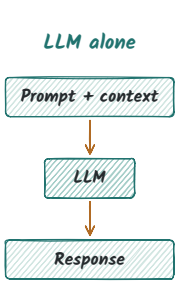
**Virtual Lab** *Nature* (2025)
LLM agents designed SARS-CoV-2 nanobodies with binding confirmed in laboratory tests.

**OpenAI: Navier–Stokes** report (2026)
LLM teams explored different approaches; the successful team ran about **10,000 agents** concurrently. **Human researchers** redirected effort and used Codex to share insights between teams.

# Agentic systems

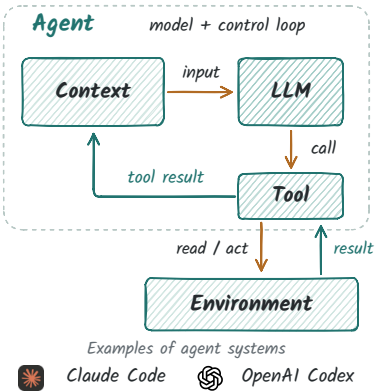
## From LLMs to research teams

# Environment, context, tools and agents




$$Y = f_\theta(p, c; \xi)$$

$p$: prompt; $c$: context
$\theta$: fixed model
$\xi$: randomness




**Environment**
The workspace an agent can access: files, data, code and external services.

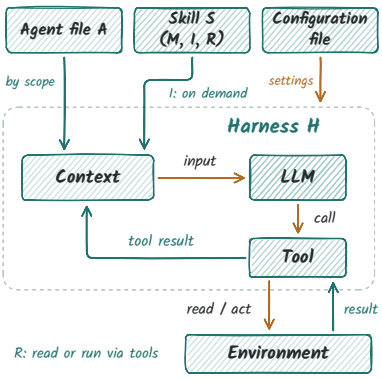
**Context**
The information given to the LLM now:
instructions, messages and tool results.

**Tool**
An interface to observe or change the environment, e.g. search or Python.

**Agent**
An LLM-based system that uses tools and chooses its next step from results.

Based on OpenAI's Codex harness documentation and Claude Code's agentic loop.

# Skills, agent files and the harness




**Agent instruction file** $A = (D, I_A)$
$D$: scope; $I_A$: standing instructions.
E.g. `AGENTS.md`: project guidance loaded into the context within its scope.

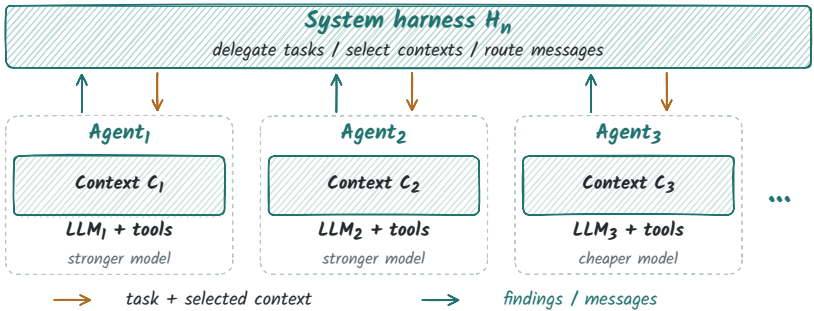
**Skill** $S = (M, I, R)$
$M$ — **metadata**: name, what the skill does, and when to use it.
$I$ — **instructions**: method, decision rules, checks and expected output.
$R$ — **resources**: optional documents, code, examples and templates.

**Harness** $H$
Orchestration software. A configuration file sets roles, models, tools, skills and routing.

Working abstractions for this talk. Implementation: skills, `AGENTS.md`, Codex harness (OpenAI Docs).
Foundations: ReAct — Yao et al. (2022), Voyager — Wang et al. (2023), SWE-agent — Yang et al. (2024).

# Multi-agent systems: who sees what?

Recall $Y = f_\theta(p, c; \xi)$. So context engineering is very important.




$H_n$ is the software coordinating $n$ agents, their contexts and communication.

A configuration file sets roles/models/tools; skills and agent files supply instructions.

Same or different models. Sources: AutoGen (Wu et al., 2023); OpenAI agent configuration.

# Why manage context and communication?

Very simply, an LLM produces random text $Y = f_\theta(p, c; \xi)$, whose distribution **depends strongly on the context**.

**1. Keep context focused** Liu et al. (2024); Modarressi et al. (2025)
**Problems:** context distraction and context clash.
Give each agent the information it needs; long histories can obscure relevant details.

**2. Give review a fresh context** Sharma et al. (2023); Dhuliawala et al. (2023)
**Problems:** context poisoning and context confusion.
Review the draft and criteria without the writer's history; verify independently.

**3. Match the model to the task** Ong et al., *RouteLLM* (2024)
Routing between stronger and cheaper models can improve the balance of cost and quality.

**4. Share findings and avoid duplicate work** Du et al. (2023); Wang et al. (2025)
Give each agent the results it needs. Redirect repeated work when it adds no new information.

Effects depend on the task and models. Fresh-context review is a design inference from bias and verification studies.

# Heuristics for agentic research

| Heuristic | Intuition |
|---|---|
| **Explore distinct approaches independently.** [1] | Different starting points can reveal approaches that others miss. |
| **Fix identified errors; stop if no progress is made.** [2] | Check that each error was repaired. Repeated approval is not evidence of a correct proof. |
| **Ask for a proof or a counterexample.** [3] | The statement you give the agent might be false. Explicitly allow it to disprove the statement instead of assuming a proof must exist. |
| **Delegate precise mathematical obligations.** [4] | When splitting the work between agents, specify exactly what each agent needs to establish and under which assumptions. |
| **Preserve each claim's status and evidence.** [5] | Reuse must not turn a conjecture or rejected argument into an "established lemma". |

Evidence-informed design heuristics: support ranges from benchmark experiments to research-system case studies.

[1] Zhu et al. (2026); [2] Zheng et al. (2026); [3] Petrov et al. (2025);
[4] Zhao et al. (2026); [5] Xu et al. (2026).

# Agentic research: a worked example

## What do we need?

- An **agent system**: in my case, Codex.
- A precise **mathematical problem statement**.
- Relevant **related work**: definitions, known results and useful methods.
- **`instruction.md`**: the task, constraints and expected output.
- **`skills.md`**: an index of reusable methods and checks.
- **`AGENTS.md`**: standing project instructions for the agents.

Here, `instruction.md` and `skills.md` are our project documents.
Each actual Codex skill has its own `SKILL.md`.

# Point processes: a Poisson example

A **simple point process** $X$ is a random set of distinct points in $\mathbb{R}^d$, finite in bounded regions.

Its count in a region $B$ is $X(B) = \#(X \cap B)$.

### Stationarity

Shifting all points leaves the distribution unchanged:

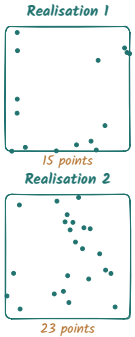
$$X + t \overset{d}{=} X \quad \text{for every } t \in \mathbb{R}^d.$$

### Homogeneous Poisson process, intensity $\lambda > 0$

$$X(B) \sim \text{Poisson}(\lambda|B|), \qquad \mathbb{E}[X(B)] = \lambda|B|.$$

Here $|B|$ is the volume of $B$; $\lambda$ is the mean number of points per unit volume.

Counts in disjoint regions are **independent**.

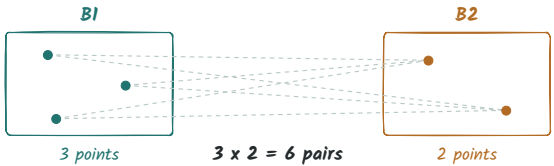



Unit-area windows; $\lambda = 20$.
On average 20 points per window.

Baddeley, Sections 1.4–1.6. Pictures show independent simulated realisations in two dimensions.

# Product densities: counting pairs

Take two **disjoint** regions $B_1, B_2$. The product $X(B_1)X(B_2)$ counts pairs with one point in each region.




Across repeated realisations, for small regions around $x_1, x_2$:

$$\underbrace{\mathbb{E}[X(B_1)X(B_2)]}_{\text{expected number of pairs}} \approx \underbrace{\rho^{(2)}(x_1, x_2)}_{\text{product density}} |B_1||B_2|.$$

**Product density:** local abundance of pairs, after accounting for the sizes of the two regions.

Three or four regions: count **triples** or **quadruples** in the same way.

The approximation applies where the density varies little across the small regions.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

# Product densities: comparison with Poisson

For a homogeneous **Poisson process**, independence gives

$$\rho^{(2)}(x_1, x_2) = \lambda^2, \qquad \rho^{(n)}(x_1, \ldots, x_n) = \lambda^n.$$

For a **stationary process** with intensity $\lambda > 0$, compare its product density with this reference:

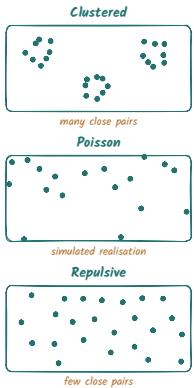
$$g_n(x_1, \ldots, x_n) = \frac{\rho^{(n)}(x_1, \ldots, x_n)}{\lambda^n}.$$

This is the *n*-**point correlation function**.

Compared with Poisson at the same intensity:

$g_n > 1$: **more such tuples.**
$g_n < 1$: **fewer such tuples.**
$g_n = 1$: the same local tuple density.

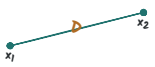
$n = 2$: pairs; $n = 3$: triangles; $n = 4$: tetrahedra.



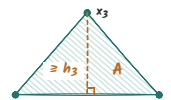

Poisson simulated; other patterns schematic. Comparisons are local; $g_2 = 1$ alone does not imply Poisson.
Baddeley, Section 2.3; Last and Penrose, Section 4.4.

# Step 1: the precise problem

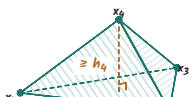
Fix $M, \lambda_0, h_3, h_4 > 0$.
Find **one stationary, simple** point process $X$ on $\mathbb{R}^3$ with intensity $\lambda_0$, finite local moments through order four and product densities.



Distance D



Area A



Volume V

For some constants $C_2, C_3, C_4 > 0$, require **simultaneously**:

$$g_2(x_1, x_2) = \frac{C_2}{D}, \qquad g_3(x_1, x_2, x_3) = \frac{C_3}{A}, \qquad g_4(x_1, x_2, x_3, x_4) = \frac{C_4}{V}.$$

Almost everywhere on nondegenerate configurations of diameter $\leq M$, with every triangle altitude $\geq h_3$ and every tetrahedron altitude $\geq h_4$.

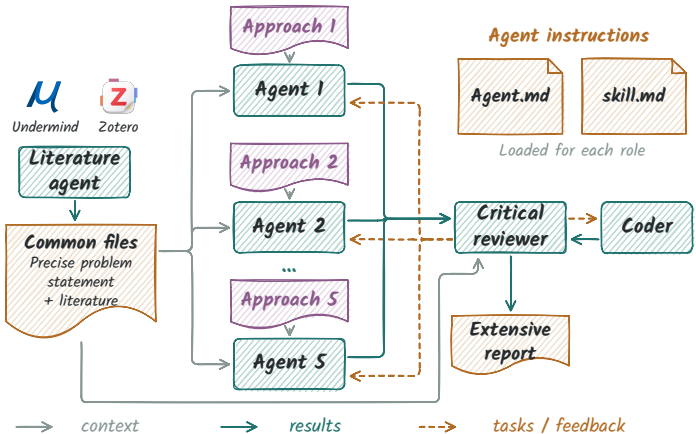
$C_2, C_3, C_4$ are chosen jointly with the process, not prescribed independently.

Current construction: sufficiently small $\lambda_0$; arbitrary prescribed intensity remains open.

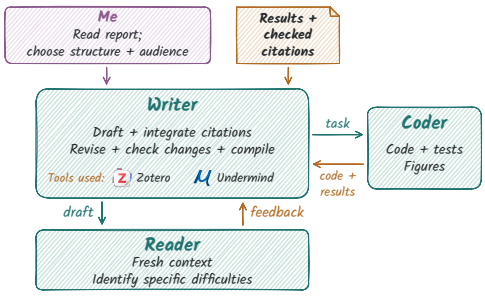
# Step 2: configuration/agent files and execution

**Research approaches**

1. Poisson cluster process
2. Cox process
3. Hawkes process
4. An approach of the agent's choice
5. Prove non-existence



# Step 3: Writing the paper




**Complete-task statistics**

| Measure | Complete task |
|---|---|
| Input tokens | 131,866,681 |
| Cached input (included above) | 128,329,472 |
| Uncached input | 3,537,209 |
| Output (includes reasoning) | 666,705 |
| Reasoning (subset of output) | 286,459 |
| **Total tokens** | **132,533,386** |
| Task-turn runtime | **3:47:06** |
| Research coordinator runtime | **3:36:43** |
| API-equivalent cost | **$78.81** |

**Writing stage: token use and cost**

| Participant | Uncached input | Cached input | Output/reasoning | Cost |
|---|---|---|---|---|
| Coordinating writer | 0.30M | 3.5M | 0.15M | $5.60 |
| Simulation specialist | 0.15M | 1.4M | 0.10M | $3.16 |
| Fresh reader | 0.06M | 0.1M | 0.02M | $0.68 |
| **Total** | **0.51M** | **5.0M** | **0.27M** | **$9.44** |

# Results and personal feedback

## Poisson cluster model

Homogeneous Poisson parents $\Pi$ each generate **0–3 children**, with jointly chosen offsets $C_z$:

$$X = \bigcup_{z \in \Pi} \big( \{z\} \cup (z + C_z) \big).$$

$C_z$ contains only child offsets. **The parent is retained**, giving 1–4 points per cluster.

**How many points?** The solution gives a probability distribution over cluster sizes $j = 1, \ldots, 4$.

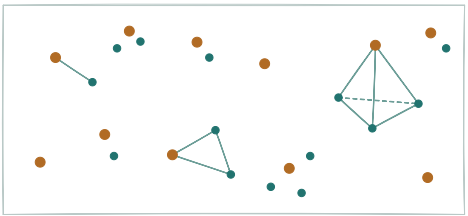
**Where are they placed?** For $j \geq 2$, the normalised density

$$\frac{F_j(z, x_1, \ldots, x_{j-1})}{Z_j}$$

describes how the $j - 1$ children are jointly distributed around the parent $z$.

## Personal feedback

| Aspect | Feedback |
|---|---|
| Mathematical correctness | Positive |
| Research and process documentation | Positive |
| Code writing | Positive |
| Text writing | Medium |
| Paper structuring | Negative |
| Mathematical explanation | Medium |



# Future of agentic research

## What I expect to change

Stronger models can increasingly plan, divide work into tasks, use tools and delegate to subagents.

For a precise mathematical problem, **one initial prompt may increasingly suffice**. This can still mean substantial autonomous work, with no guarantee of success.

As platforms provide more harness functions, we will have **less to build ourselves**.

I also expect explanations to become **less sloppy and more useful for learning**.

## Where a custom harness still helps

- Keep the exact target, hypotheses and criteria for success fixed.
- Require separate review and formal verification.
- Keep human checkpoints for direction and interpretation.
- Record results, failed approaches, sources and dependencies.
- Set compute budgets, stopping rules and model choice.
- When writing papers with AI, set your preferences for writing and illustration styles.

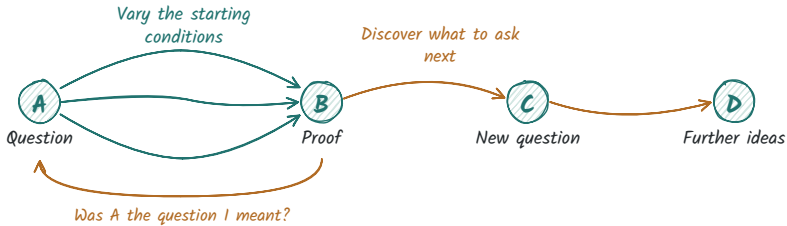
**A reported example.** For its October release, OpenAI says almost all results came from one prompt to one agent. Some may have needed retries; the full workflow has not been independently reproduced.

Scientific American, October 2026; OpenAI release details.

# Personal take

I find this exciting: an elegant solution, with reasoning that feels human.
Still, something is missing when I do not discover the ideas myself.

**Faster proofs let me explore more. Research was never a straight line.**




**Reaching B only helps me if I understand the mathematics.**

**My concern as a PhD student:** outsourcing too much may bypass the learning.
A PhD is still an education. Producing results and developing understanding can come apart.

Related: *A Severe Misalignment of AI in Mathematics* (joint declaration, 11 September 2026).

# Thank you!

## Questions and discussion

Slides and, later, the experiments:

github.com/MathisRost/agentic-ai-math

